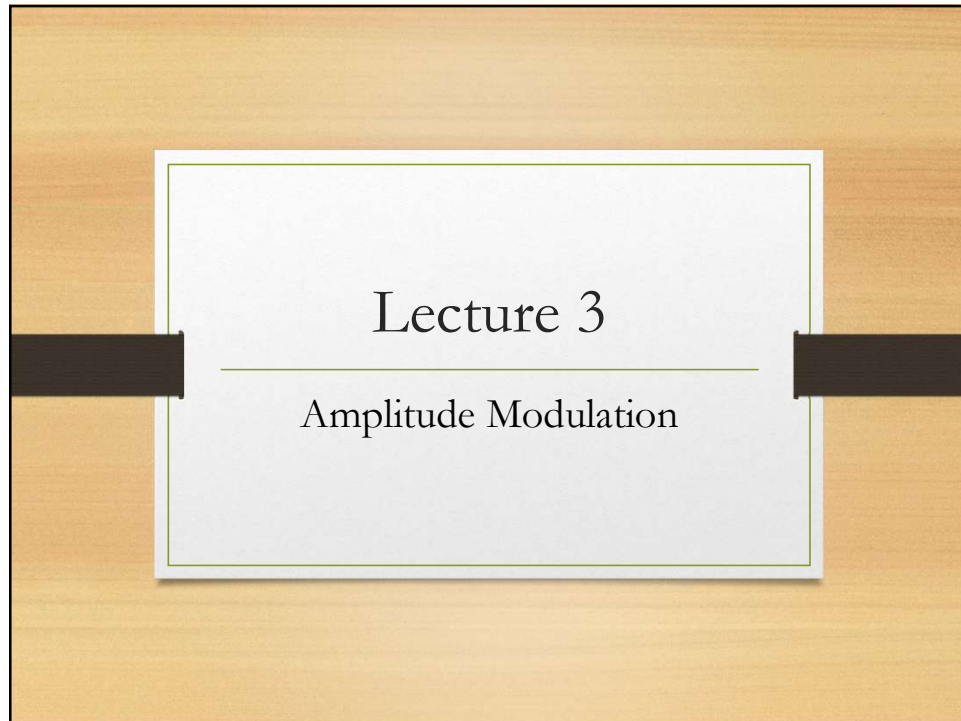
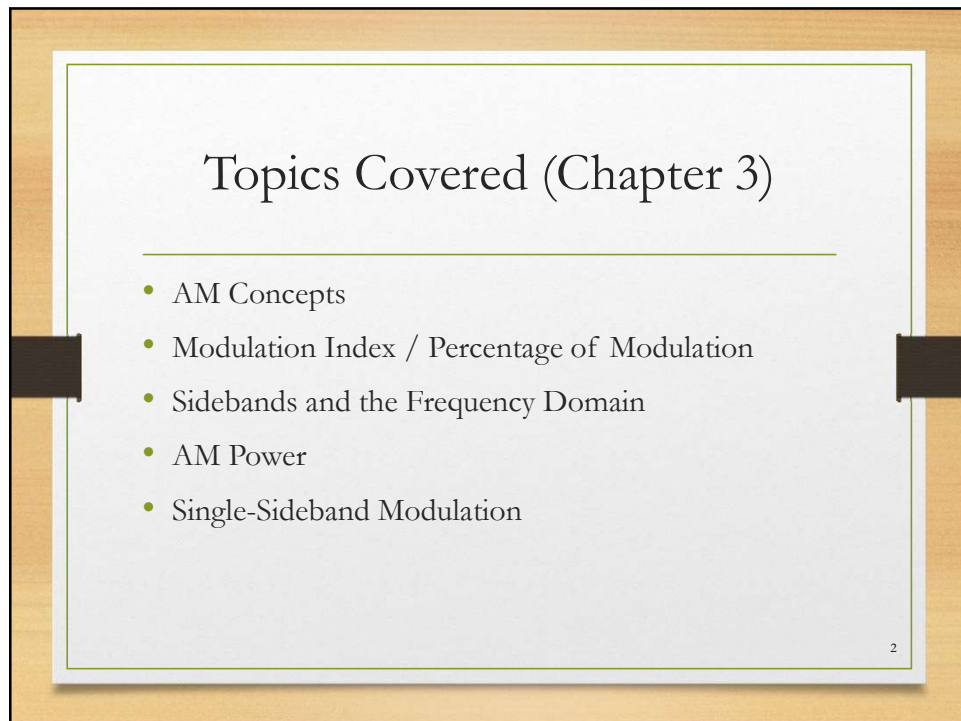




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3-1: AM Concepts

- In the modulation process, the voice, video, or digital signal modifies another signal called the carrier.
- In **amplitude modulation (AM)** the information signal varies the amplitude of the carrier sine wave.
- The instantaneous value of the carrier amplitude changes in accordance with the amplitude and frequency variations of the modulating signal.
- An imaginary line called the **envelope** connects the positive and negative peaks of the carrier waveform.

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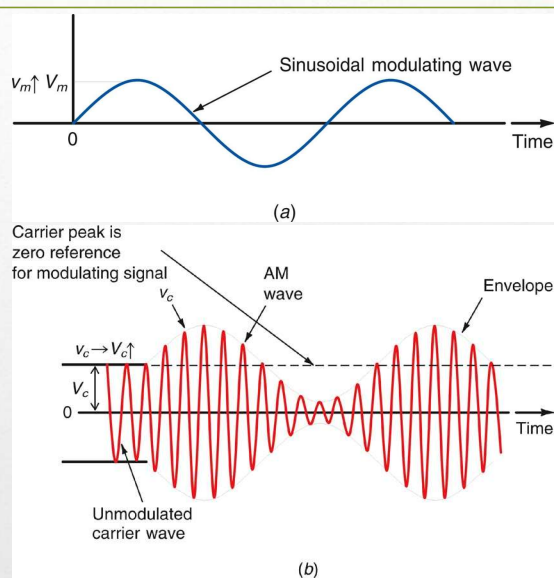


Figure 3-1: (a) modulating signal. (b) The modulated carrier.

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AM Signal

- For a single-tone amplitude modulated signal (message signal $V_m \cos(2\pi f_m t)$), it can be mathematically described as

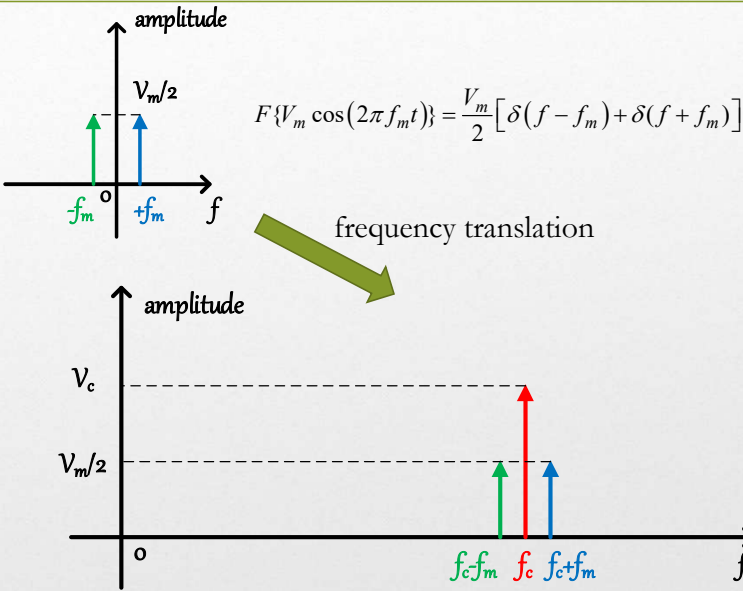
$$v_{AM}(t) = [V_c + V_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

- Furthermore, we have

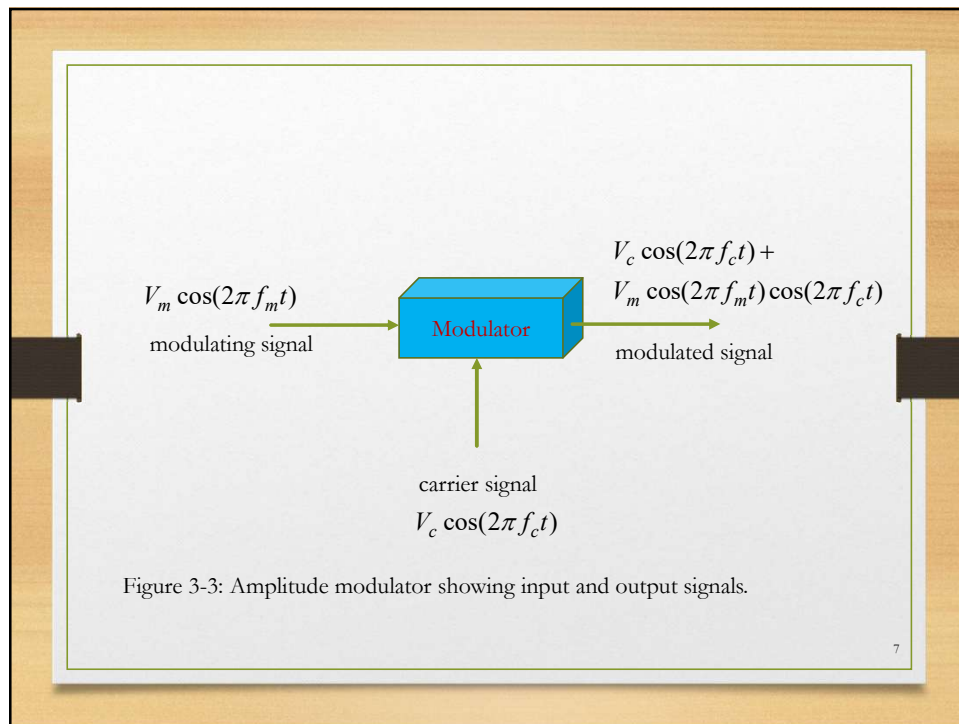
$$\begin{aligned} v_{AM}(t) &= V_c \cos(2\pi f_c t) + V_m \cos(2\pi f_m t) \cos(2\pi f_c t) \\ &= \underbrace{V_c \cos(2\pi f_c t)}_{\text{carrier}} + \underbrace{\frac{V_m}{2} \cos[2\pi(f_c - f_m)t]}_{\text{lower-side band}} + \underbrace{\frac{V_m}{2} \cos[2\pi(f_c + f_m)t]}_{\text{upper-side band}} \end{aligned}$$

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Modulation Index

- The **modulation index (m)** is a value that describes the relationship between the amplitude of the modulating signal and the amplitude of the carrier signal.

$$m = V_m / V_c$$

- This index is also known as the **modulating factor** or **coefficient**, or the **degree of modulation**.
- Multiplying the modulation index by 100 gives the percentage of modulation.

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Overmodulation and Distortion

- m should be a number between 0 and 1.
- If m is greater than 1, distortion will happen.
- If the distortion is great enough, the intelligence signal becomes unintelligible.
- Distortion of voice transmissions produces garbled, harsh, or unnatural sounds in the speaker.
- Distortion of video signals produces a scrambled and inaccurate picture on a TV screen.

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3-2: Modulation Index and Percentage of Modulation

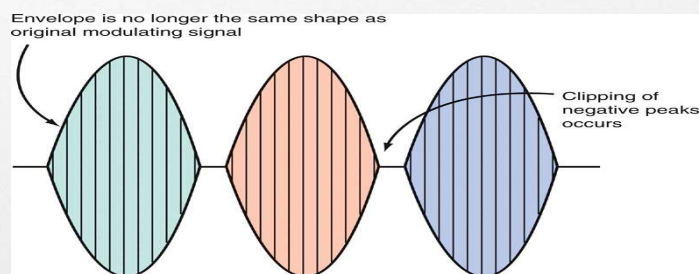


Figure 3-4: Distortion of the envelope caused by overmodulation where the modulating signal amplitude V_m is greater than the carrier signal V_c .

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3-2: Modulation Index and Percentage of Modulation

Percentage of Modulation

- The modulation index is commonly computed from measurements taken on the composite modulated waveform.
- Using oscilloscope voltage values:

$$V_m = \frac{V_{\max} - V_{\min}}{2}$$

- The amount, or depth, of AM is then expressed as the percentage of modulation ($100 \times m$) rather than as a fraction.

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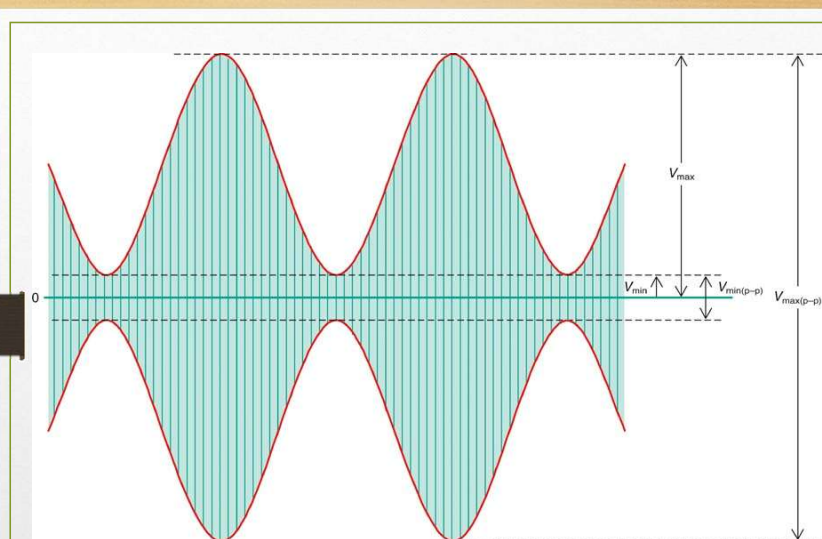


Figure 3-5: AM wave showing peaks ($V_{\max}$) and troughs ($V_{\min}$).

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3-3: Sidebands and the Frequency Domain

- Single-frequency sine-wave modulation generates two sidebands.
- Complex wave (e.g. voice or video) modulation generates a range of sidebands.
- The upper sideband (f_{USB}) and the lower sideband (f_{LSB}) are calculated:

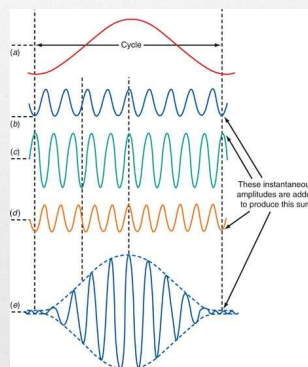
$$f_{\text{USB}} = f_c + f_m \quad \text{and} \quad f_{\text{LSB}} = f_c - f_m$$

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3-3: Sidebands and the Frequency Domain

Figure 3-6: The AM wave is the algebraic sum of the carrier and upper and lower sideband sine waves. (a) modulating signal. (b) Lower sideband. (c) Carrier. (d) Upper sideband. (e) Composite AM wave.



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3-3: Sidebands and the Frequency Domain

Frequency-Domain Representation of AM

- Observing an AM signal on an oscilloscope, you see only amplitude variations of the carrier with respect to time.
- A plot of signal amplitude versus frequency is referred to as **amplitude spectrum** (frequency-domain display).
- A spectrum analyzer is used to display the frequency domain as a signal.
- Bandwidth is the difference between the upper and lower sideband frequencies.

$$BW = f_{USB} - f_{LSB}$$

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3-3: Sidebands and the Frequency Domain

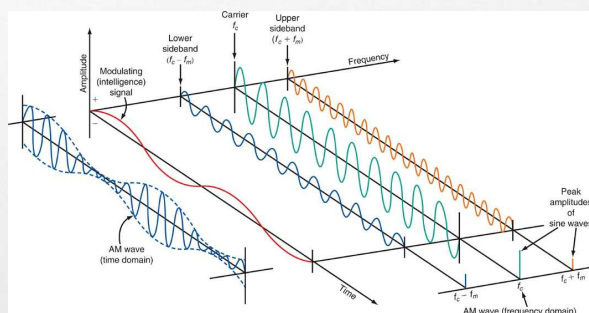


Figure 3-8: The relationship between the time and frequency domains.

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3-3: Sidebands and the Frequency Domain

Frequency-Domain Representation of AM

- Example:**

A standard AM broadcast station is allowed to transmit modulating frequencies up to 5 kHz. If the AM station is transmitting on a frequency of 980 kHz, what are sideband frequencies and total bandwidth?

$$f_{\text{USB}} = 980 + 5 = 985 \text{ kHz}$$

$$f_{\text{LSB}} = 980 - 5 = 975 \text{ kHz}$$

$$\text{BW} = f_{\text{USB}} - f_{\text{LSB}} = 985 - 975 = 10 \text{ kHz}$$

$$\text{BW} = 2 (5 \text{ kHz}) = 10 \text{ kHz}$$

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3-4: AM Power

- In radio transmission, the AM signal is amplified by a power amplifier.
- A radio antenna has a characteristic impedance that is ideally almost pure resistance.
- The AM signal is a composite of the carrier and sideband signal voltages.
- Each signal produces power in the antenna.
- Total transmitted power (P_T) is the sum of carrier power (P_c) and power of the two sidebands (P_{USB} and P_{LSB}).

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3-4: AM Power

- Output power can be calculated by using the formula

$$P_T = (V_T)^2 / R$$

where V_T is RMS output voltage and R is resistive load impedance

- For single-tone full AM modulation

$$P_{LSB} = P_{USB} = \frac{\left(\frac{V_m}{2\sqrt{2}}\right)^2}{R} = \frac{V_m^2}{8R} = \frac{(mV_c)^2}{8R} = \frac{m^2}{4} \times \frac{V_c^2}{2R} = \frac{m^2}{4} P_c$$

$$P_{AM} = P_c + P_{LSB} + P_{USB} = P_c + 2 \times \frac{m^2}{4} P_c = P_c \left(1 + \frac{m^2}{2}\right)$$

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3-4: AM Power

- The greater the percentage of modulation, the higher the sideband power and the higher the total power transmitted.
- Maximum power appears in the sidebands when the carrier is 100 percent modulated.
- For an arbitrary modulating signal, the modulation index can be calculated as

$$m = \sqrt{\frac{2P_{SB}}{P_c}}$$

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3-5: Single-Sideband Modulation

- In amplitude modulation, two-thirds of the transmitted power is in the carrier, which conveys no information.
- Signal information is contained within the sidebands.
- **Single-sideband (SSB)** is a form of AM where the carrier is suppressed and one sideband is eliminated.

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DSSC Signal

- The first step in generating an SSB signal is to suppress the carrier, leaving the upper and lower sidebands.
- This type of signal is called a **double-sideband suppressed carrier (DSSC)** signal. No power is wasted on the carrier.
- A balanced modulator is a circuit used to produce the DSSC signal's sum and difference frequencies but to cancel or balance out the carrier.
- DSB is not widely used because the signal is difficult to demodulate (recover) at the receiver.

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DSSC Signal

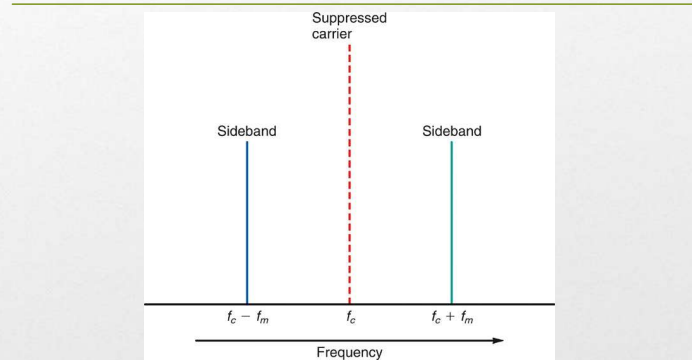


Figure 3-16: A frequency-domain display of DSB signal.

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3-5: Single-Sideband Modulation

- One sideband is all that is necessary to convey information in a signal. A **single-sideband suppressed carrier (SSSC)** signal is generated by suppressing the carrier and one sideband.
- SSB signals offer four major benefits:
 1. Spectrum space is conserved and allows more signals to be transmitted in the same frequency range.
 2. All power is channeled into a single sideband. This produces a stronger signal that will carry farther and will be more reliably received at greater distances.
 3. Occupied bandwidth space is narrower and noise in the signal is reduced.
 4. There is less selective fading over long distances.

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3-5: Single-Sideband Modulation

Disadvantages of DSB and SSB

- Single and double-sideband are not widely used because the signals are difficult to recover (i.e. demodulate) at the receiver.
- A low power, **pilot carrier** is sometimes transmitted along with sidebands in order to more easily recover the signal at the receiver.

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3-5: Single-Sideband Modulation

Signal Power Considerations

- In SSB, the transmitter output is expressed in terms of **peak envelope power (PEP)**, the maximum power produced on voice amplitude peaks.

Applications of DSB and SSB

- A **vestigial sideband signal (VSB)** is produced by partially suppressing the lower sideband. This kind of signal is used in TV transmission.

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